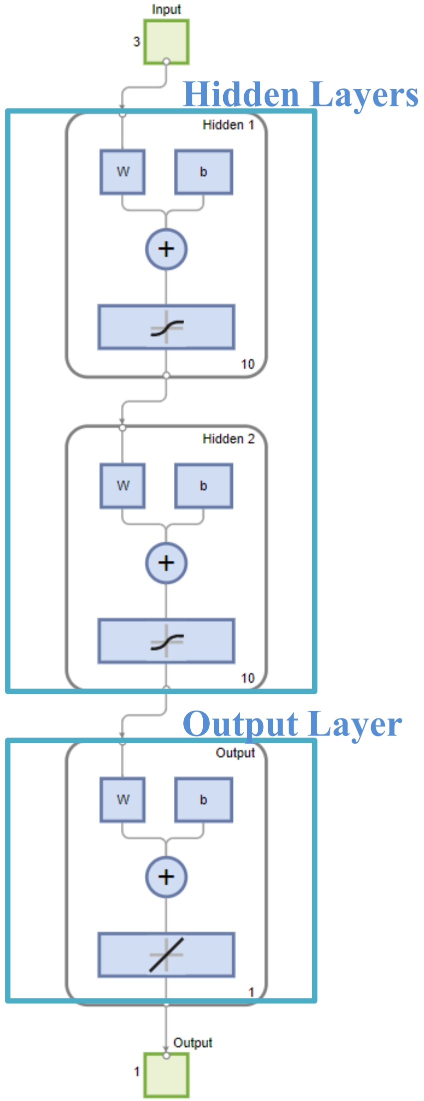
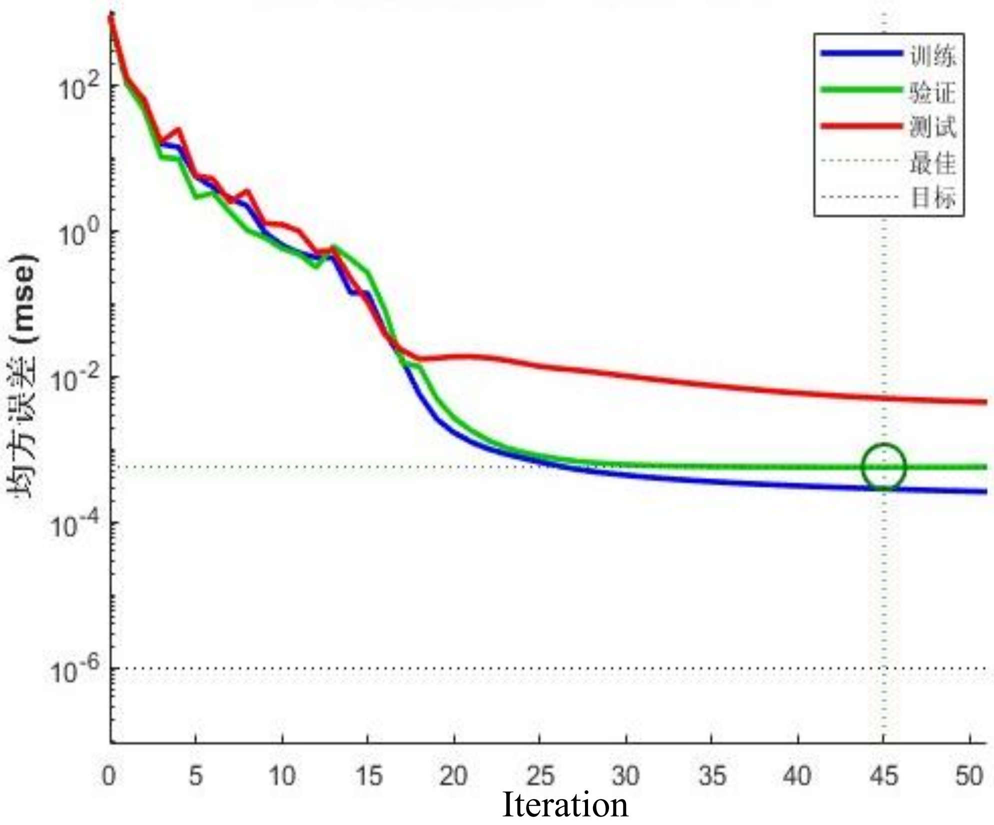
Part 2: Optimization of DRA Using PSO with NN as a Surrogate Model

2.1 Surrogate Model

NN Architecture



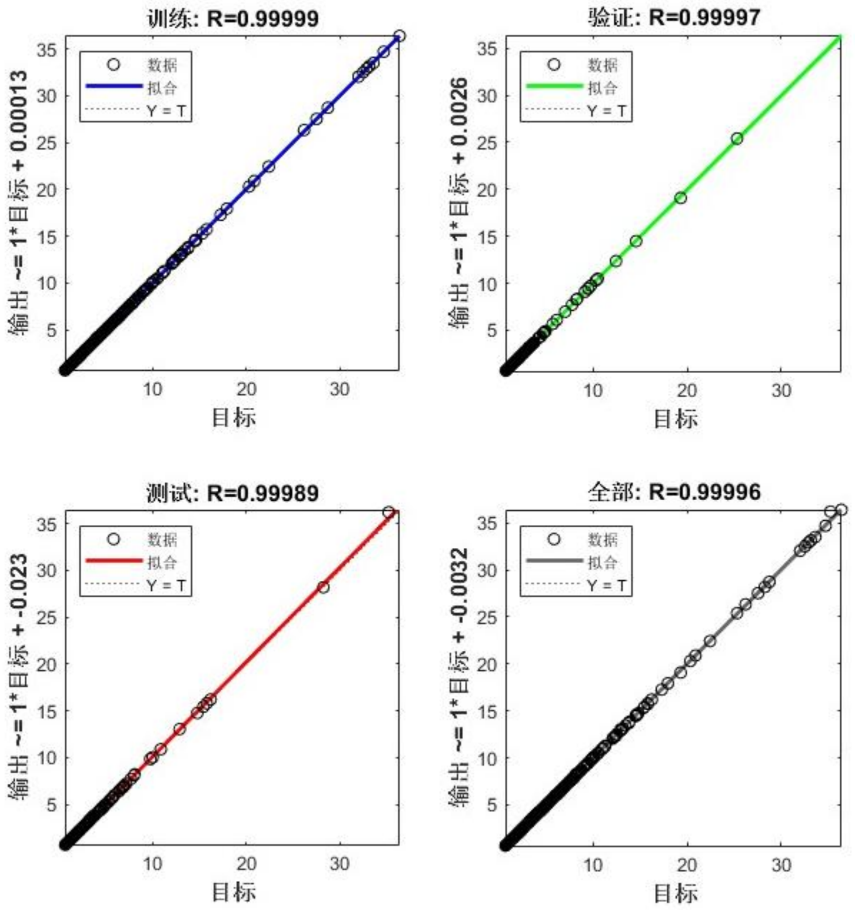
Loss Function Convergence



Training Time: 0.987894 s

MSE on Test dataset: 0.0091

Regression Analysis of BPNN Predictions



The neural network predicts well on training, validation, and test sets with R-values near 0.999, showing strong accuracy and generalization.

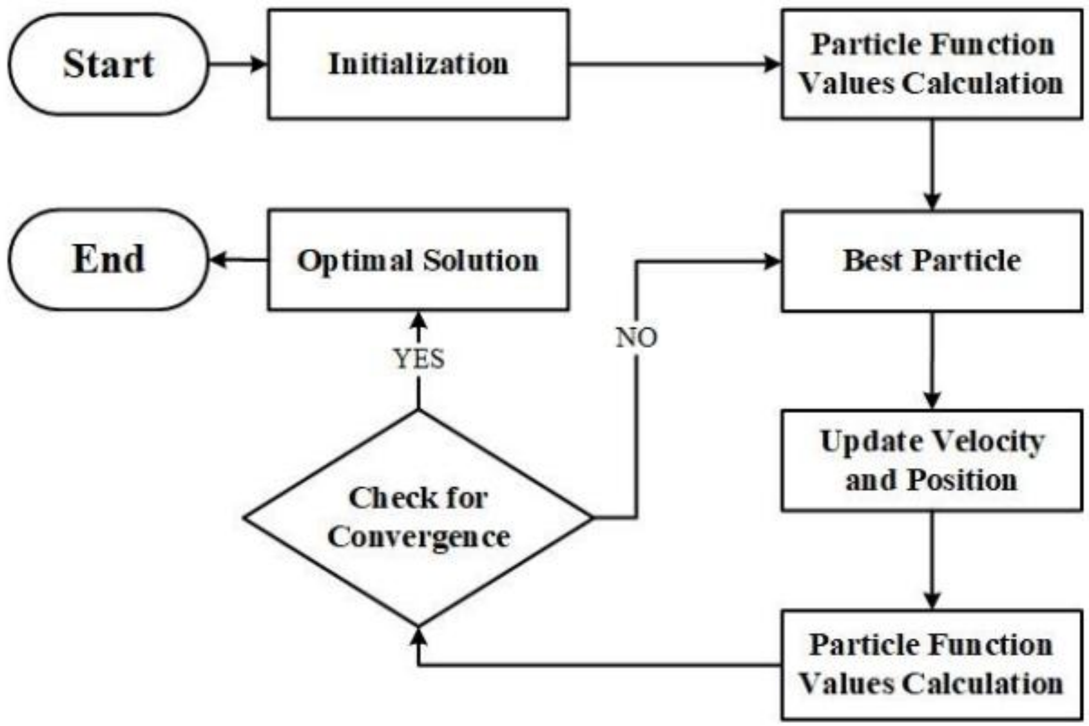
Total parameters = 40+110+11=**161**

Executed on a system equipped with an AMD Ryzen 7 7840H CPU with integrated Radeon 780M Graphics and an NVIDIA GeForce RTX 4060 Laptop GPU.

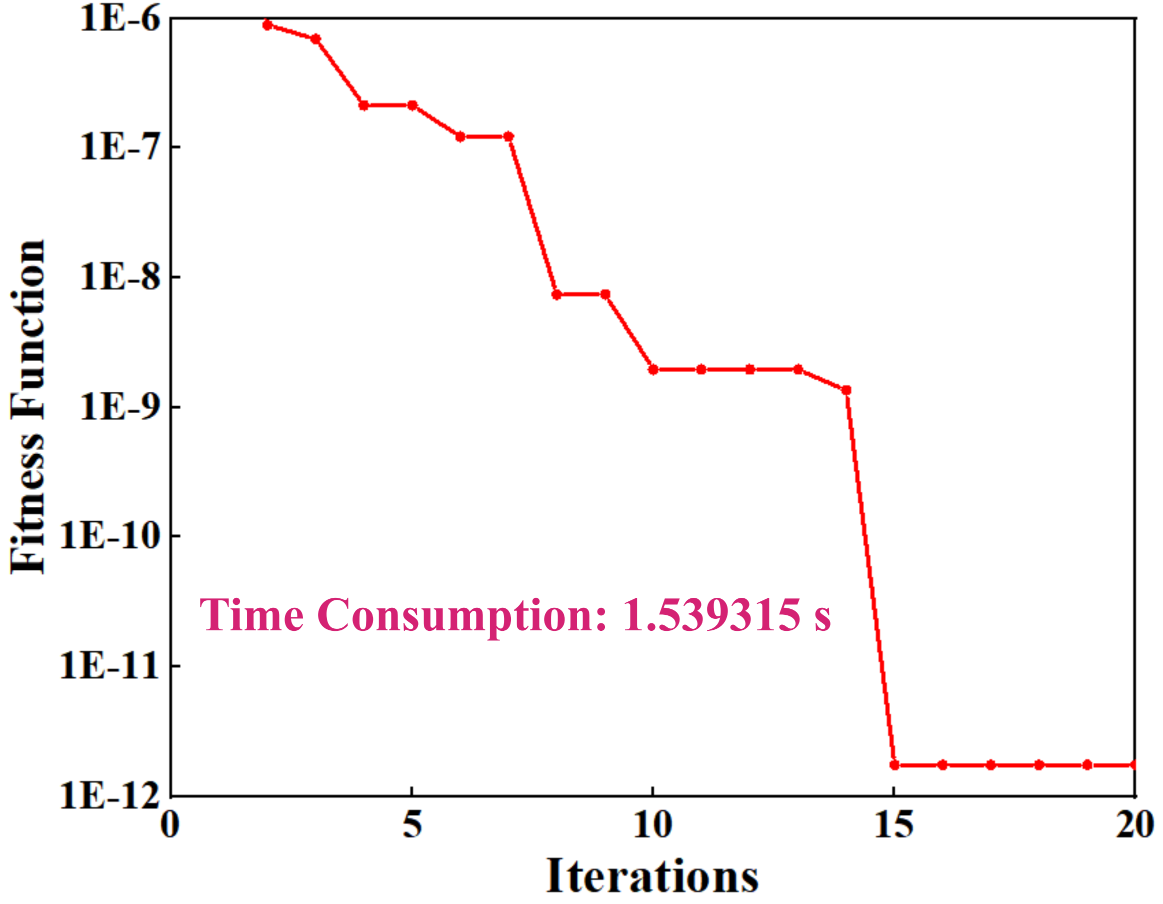
Part 2: Optimization of DRA Using PSO with NN as a Surrogate Model

2.2 PSO Process and Results

Flowchart of PSO



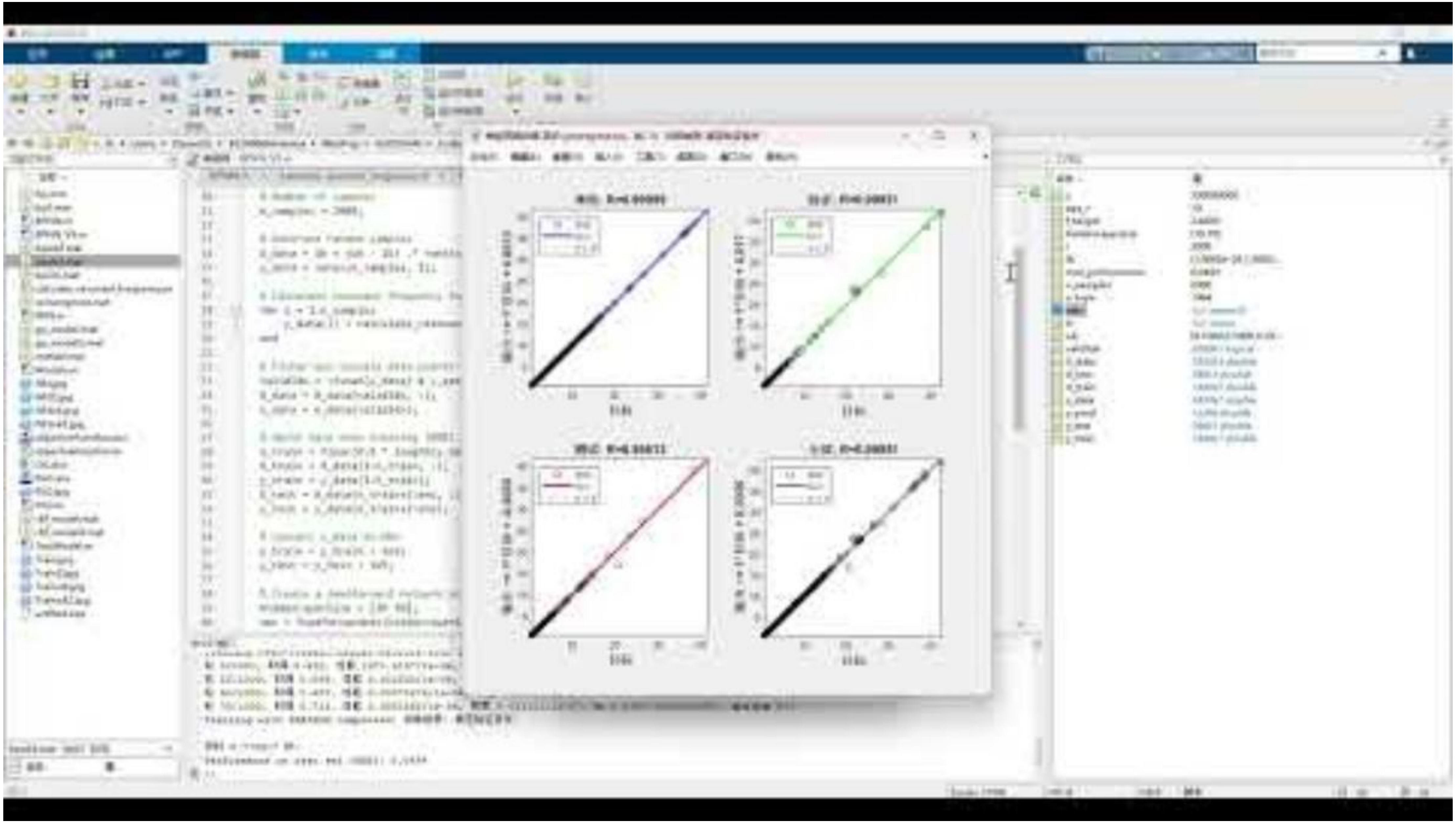
PSO Convergence



$$\boldsymbol{Fit}(a, b, d) = \begin{cases} \min\limits_{a,b,d \in D} \{abd + \alpha|f - f^*|\}, if\ |f - f^*| < 0.1\ GHz \\ 1, \qquad\qquad Otherwise \end{cases}$$

a (mm)	b (mm)	d (mm)	v (mm^3)	f (GHz)	f_{error} (GHz)
48.772	17.104	12.718	10690	2.399998	0.000002

Part 2: Optimization of RDRA by PSO Used NN as a Surrogate Model



Demo Video